

External Adversarial Audit Report — Paper III (v0.9.5)

Linear-Error Clique Partitions of Split Graphs (Erdős #81, Paper III), review version v0.9.5. Engagement per MANIFEST0/AUDIT_MANDATE.md: adversarial stance, Lean 4 formalization **out of scope**, verdicts earned by reproducible evidence only.

Auditor: external adversarial audit agent (Claude, Anthropic model `claude-fable-5`), 2026-07-21. Package audited: SHA-256 record in `received_inputs.sha256` (56 files; all 54 entries of the package's own manifest verified bit-identical on receipt).

1. Executive verdict

PASS_WITH_OBSERVATIONS

No counterexample, no overstated axiom, no circular or missing dependency, no statement $\leftrightarrow$ claim mismatch, and no arithmetic error in a load-bearing step was found — under adversarial testing that exceeded the internal audit's ranges by one to seven orders of magnitude and used independent methods throughout. The observations are: one minor overstatement of "self-containment" (classical Dirac/Turán/ $\chi(K_t)$ inputs are also load-bearing but unproved in the paper), three minor presentational issues, and three minor script defects in the internal audit (none verdict-affecting).

Scope reminder: Theorem 1.1 and Corollary 1.2 are *conditional on AX1 and AX2* (verified faithful to the published literature, not stronger). Their unconditional content — Proposition 10.1, Theorems 3.1/4.2, Lemmas 4.1–7.1, Appendices B/D — is what a finite audit can attack, and all of it survived. A computational audit cannot *prove* the for all -statements; it can only fail to break them while verifying every finite/closed-form component exactly. Qualified human expert review of the two asymptotic inputs' journal texts and of the global assembly remains advisable before publication (see §5).

2. Per-claim verdict table

Claim	Statement	Verdict	Evidence
C-1 · Theorem 1.1	$\Phi(G) \leq n^2/6 + Cn$ (relative to AX1, AX2)	VERIFIED_WITHIN_SCOPE (conditional) — assembly reconstructed, acyclic, every finite ingredient verified; asymptotic inputs faithful	Blocks A2, B, C, D
C-2 · Corollary 1.2	$cp(G) \leq n^2/6 + Cn$	VERIFIED_WITHIN_SCOPE (conditional) — follows from (1.1), which is immediate	Block A1
C-3 · Prop 10.1 (i)	$p \geq 36, 0 \leq s \leq 6\sqrt{p} \Rightarrow \Phi \leq n^2/6 + 2n$	VERIFIED_WITHIN_SCOPE — analytic chain machine-verified exhaustively for $36 \leq p \leq 600$ (D20) + instance certificates at $p \leq 100$ (C4)	D20, C4
C-3 · Prop 10.1 (ii)	$p \geq 2304, 6\sqrt{p} \leq s \leq p/8, \text{deg cond} \Rightarrow \Phi \leq n^2/6$	VERIFIED_WITH_RESIDUAL_RISK — every inequality of the §9.3 chain re-derived and grid-verified (D11–D18); true-scale instance certificates at $p=2304, s=288$ (2.44M-triangle packings, exact counters); residual: instance testing is existential, the for all -claim rests on the (verified) analytic chain	D11–D18, C4
C-4 · Theorem 3.1	$v_3^*(H(p,q,d)) = F(p,q,d)$	VERIFIED_WITHIN_SCOPE — exact rational sandwich certificates (primal orbit LP + feasible dual covers) on 49,079 instances incl. p up to 10^4 , q up to $4p$; independent raw exact rational simplex (no orbit reduction) on 266 small instances: all match. 0 failures / 49,345 total	C1, D02/D03

Claim	Statement	Verdict	Evidence
C-5 · Lemma 4.1	$v_3^*(G) \geq (1/q)\Sigma F(p,q,d_i)$	VERIFIED_WITHIN_SCOPE — exact simplex on 153 arbitrary-profile graphs, 0 violations (78 tight)	C7
C-6 · Theorem 4.2	$v_3^* \geq T + \mu(\alpha)p^2 - p/4$; core (4.5)	VERIFIED_WITHIN_SCOPE — 2,497,464 exact-integer checks (exhaustive $p \leq 150$; random $p \leq 10^9$), all three branches; residuals re-derived symbolically incl. exact square completions (D04–D07)	C2, D
C-7 · Lemma 5.1 + Cor 5.3	$v_3 \geq (1/q)\Sigma C(d_i, 2)$; $\Phi \leq n^2/6 + p/2 + (s^2 - 6s + 3)/12$	VERIFIED_WITHIN_SCOPE — 443 applicable instances confirmed by packing certificates + 40-instance exact- v_3 sample; (5.2)/(5.3) algebra proved as identities (D09/D10)	C3, D
C-8 · Lemma 5.2	double-factor inequality (5.4)	VERIFIED_WITHIN_SCOPE — 443 applicable instances (certificates + exact sample); V computed exactly per instance	C3
C-9 · Lemma 6.1	polarization $V \geq ((2p - 3m - 1)/4)\Sigma \setminus$	$S_i \Delta S_j \setminus$	
C-10 · Lemma 7.1	reserved-gain shifted-center (7.6)	VERIFIED_WITHIN_SCOPE — 525 applicable (R, instance) pairs confirmed (hypotheses (7.1)–(7.2) checked per instance); assembly identity (7.6) reconstructed and machine-verified (D23 — the paper does not display it)	C3, D23
C-11 · Appendix B	$\text{Odd}(J) = O, \setminus$	$E(J) \setminus$	$\leq p - 1, \Delta(J) \leq 2$
C-12 · Appendix D	self-contained list edge coloring	VERIFIED_WITHIN_SCOPE (logical audit) — D.1–D.3 read line-by-line: kernel lemma, Gale–Shapley, inline König (bipartite odd-path parity correct), out-degree $\leq \Delta - 1$, kernels=stable matchings; hypothesis check $\setminus$	L\
C-13 · Sharpness §10.2	$K_{pv}K_{2p}$ attains $n^2/6 + n/6$	VERIFIED_WITHIN_SCOPE — $v_3 = C(p, 2)$ certified both directions, $p \leq 24$; closed form is a polynomial identity (D22); no audited instance beat 1/6	C3/C5, D22
C-14/15/16 · Addenda	Cor 10.4 / 10.4b / 12.2	VERIFIED_WITHIN_SCOPE — restatements/specializations of C-4/C-1/C-3; no new mathematical content; Cor 10.4's extended range ($q \geq 0$, all d) explicitly covered by C1's grid	C1, A1
AX1 · Theorem 2.1	$v_3^* - v_3 = o(\setminus)$	$V \setminus$	$^2)$ uniform
AX2 · Theorem 2.3	triangle-divisible, $\delta \geq (0.9 + \epsilon) \setminus$	$V \setminus$	$\Rightarrow$ decomposition

3. Findings

Blocking: none. Major: none. Minor: 6.

1. **A-04 (minor).** §2.4's "Sections 5–7 and Proposition 10.1 depend on no external theorem" and §11.3's "no external theorem whatsoever" overstate: the classical 1-factorization $\chi'(K_t)$ (eq. 2.2, stated without proof) is load-bearing for Lemmas 5.1/5.2/7.1 and hence Prop 10.1; §8 additionally uses Dirac and Turán. These are classical, constructive facts (this audit constructs the factorizations explicitly), and LEDGER.md lists them openly — but the wording should say "no external theorem beyond classical finite facts ($\chi'(K_t)$, Dirac, Turán)".

2. **A-05 (minor, presentational).** Figure links in the review md point to a directory not shipped at that relative path.

3. **A-06 (minor, reproducibility).** §11.6's "46,390 + 91 checks" and Appendix C's script names are not reproducible from this package (different audit generation shipped). No logical weight (the paper states no computation is a premise).

4. **E-02 (minor).** Internal block02's docstring advertises an "(EXACT)" second verification that is not implemented.

5. **E-03 (minor).** Internal block02 is float-with-tolerance ($1e-7$) by design — masking-capable; did not bite (this audit's exact certificates agree).

6. **E-04 (minor).** Internal block04 ignores the CBC solver status; fail-safe direction only (cannot create a false PASS). Status-checked re-runs agree 40/40 with this audit's independent branch-and-bound.

Corrections against the auditor (recorded symmetrically): two initial defects in this audit's own Block D script (wrong guessed square-completion center; malformed D16 check) were found and fixed during development; one Block C4 certificate construction (mixed two-center) was initially too weak and produced a spurious FAIL that was diagnosed as an auditor-side construction deficit, not a paper defect, and repaired following the paper's own mechanism (after which it PASSED with slack 1,397).

4. What was attacked and how (summary)

- **Block A:** both documents read in full; 18 ledger nodes compared statement-by-statement; DAG rebuilt from the proofs; 35 mechanical anchors; external-theorem census with line numbers. No mismatch; no circularity (§8's degree bound is hypothesis-passing, not circular).

- **Block B:** AX1/AX2 checked against primary arXiv statements (verbatim sentences located); citation metadata verified; usage localization confirmed mechanically.

- **Block C:** ~2.61 million adversarial instances, zero violations:
C1 49,345 certified equalities $v_3^*=F$ (two independent exact methods: 49,079 sandwich certificates + 266 raw exact simplex); C2 2,497,464 margin checks ($p \leq 150$ exhaustive, $p \leq 10^9$ random, all-integer); C3 821 instances / 1,854 bound checks with verified packing certificates + forced exact- v_3 refutation sample; C4 true-scale corridor certificates incl. $p=2304$ (window degeneracy [288,288] at threshold confirmed); C5 sharpness both directions; C6 Appendix B exhaustive; C7 Lemma 4.1 exact simplex.

- **Block D:** 38/38 algebra checks with a self-written exact polynomial engine (no CAS), including certificates the paper leaves implicit (branch-3 square completion with boundary case; Lemma 7.1 assembly identity D23).

- **Block E:** internal audit reproduced bit-identically (4/4 blocks); its scripts read adversarially (3 minor defects); its shared formulas stress-tested at boundaries (0 mismatches on 20k+ cases).

5. Coverage statement (honest boundary)

Tested:

- Every finite/closed-form claim (C-4...C-13) with exact arithmetic, at ranges exceeding the internal audit's: $p \leq 150$ exhaustive for (4.5) (internal: $p \leq 48$); $p \leq 40$ full grid + $p \leq 10^4$ sampled for $v_3^*=F$ with q up to $4p$ (internal: $p \leq 8$, $q \leq 8$, float); instance families with 9 adversarial profile classes (internal: 3).

- The §9.3/§10.5 corridor chains: every inequality re-derived independently and

machine-verified, including exact grids at $p=2304\dots2339$ and up to 10^5 (D15).

- Prop 10.1 at true scale: explicit packings at $p=2304$ ($n=6624$, up to 12.2M edges), edge-disjointness verified by exact counters.

- AX1/AX2 faithfulness at the primary-abstract level; usage localization.
- The internal audit: full re-run + code review + boundary stress.

NOT tested / limitations:

- The Lean 4 formalization — **excluded by mandate**; nothing here relies on it.
- The truth of AX1/AX2 themselves — axioms by engagement design; only faithfulness

to the literature was audited, and for Haxell–Rödl (Combinatorica) and BKLO (Adv. Math.) the paywalled journal texts were verified at abstract/secondary level only (the two decisive sentences appear verbatim in the Yuster and Dross arXiv abstracts, which are primary).

- for all -claims over infinite families (Thm 1.1, Prop 10.1, lemmas at all orders) cannot

be established by computation. For these, what was verified is: (a) the complete analytic chains, step-by-step, in exact arithmetic; (b) adversarial instances, which all confirmed. The global Theorem 1.1 additionally rests on the two asymptotic inputs and on the regime-assembly logic (verified by reading, A2).

- Appendix D was audited logically (line-by-line), not computationally.
- Part (ii) instance certificates used common/two-center profiles; fully generic

profiles at $p=2304$ (requiring the IRQ list-coloring family) were not instantiated — the analytic chain covering them is verified (D15–D18, C3's Lemma 7.1 tests at small scale exercise IRQ hypotheses (7.1)–(7.2) directly).

- No independent literature/novelty search was performed (out of the mandate's scope).

Residual risks: LOW overall. The single locus of non-elementary trust is the two asymptotic inputs (paywalled full texts; abstract-level verification) — flagged VERIFIED_WITH_RESIDUAL_RISK at the source-inspection level, standard for these well-known results.

6. Recommendation

Maintain the paper's claims as stated, with the wording fix of Finding A-04 (list the classical inputs explicitly), the presentational fixes A-05/A-06, and the internal- audit script repairs E-02/E-03/E-04. A qualified human review of the two paywalled journal statements (Combinatorica 21:13–38; Adv. Math. 288:337–385) before the v1.0 preprint would close the last source-verification gap. This report does not modify any authoritative gate record.

7. Reproduction appendix

Environment: ENVIRONMENT.md. All scripts are deterministic (fixed seeds), write full logs to their block's results/, and exit nonzero on any failure.

```
python blockA_faithfulness/check_census.py
python blockC_counterexample_search/c1_nu3star_certificates.py      # ~1-2 h
python blockC_counterexample_search/c2_margin_grid.py              # ~5 s
python blockC_counterexample_search/c3_lemmas_packing.py           # ~30 s
python blockC_counterexample_search/c4_corridor.py                  # ~1 min
python blockC_counterexample_search/c6_appendixB.py                 # ~1 s
python blockC_counterexample_search/c7_cloning_lemma41.py          # ~10 s
python blockD_algebra_rederivation/rederive_algebra.py             # ~30 s
python blockE_audit_the_audit/e3_boundary_stress.py                 # ~30 s
python build_deliverable.py                                          # certificates+ziips+manifest
```

Internal-audit re-run (Block E1): copy OUR_INTERNAL_AUDIT/, move each block's results/ aside, run the four verify_*.py, diff outputs (no timestamps inside).